Identification, Summary Design, and Method Comparison in Simulation-Based Inference for Adaptive-Network Epidemics

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Abstract

We study simulation-based inference for a stochastic SIR epidemic evolving on an adaptive contact network with transmission probability β, recovery probability γ, and rewiring probability ρ. The likelihood is intractable because infection, recovery, and rewiring interact through an unobserved network whose topology changes endogenously over time. The central challenge in this problem is not only computation but identification: infected-trajectory summaries alone cannot cleanly separate transmission, recovery, and network adaptation. I therefore compare rejection ABC, regression-adjusted ABC, ABC-MCMC, SMC-ABC, and synthetic likelihood using shared simulations and three nested summary-statistic sets. The main empirical result is that rewiring summaries are essential for identifying ρ, while final-degree summaries provide additional refinement. Among the methods implemented here, regression-adjusted ABC on the full summary set gives the strongest overall posterior predictive performance on the observed data. A synthetic-truth recovery exercise shows that both adjusted ABC and synthetic likelihood recover a chosen ground-truth parameter vector in this run, but synthetic likelihood achieves this with a much larger simulation budget and still exhibits a very tight local β–ρ dependence on the actual data. The broader lesson is that in this model, summary design matters at least as much as algorithm choice, and advanced SBI methods should be judged by fit, calibration, robustness, and computational cost rather than posterior sharpness alone.

1 Introduction

When an epidemic spreads through a contact network, infection risk depends on who is connected to whom. In this project, the challenge is not only that the network is unobserved, but also that it changes during the outbreak because susceptible agents can sever links to infected neighbors and reconnect elsewhere. This creates a feedback loop between epidemic dynamics and network structure, making the model scientifically meaningful but statistically difficult. The resulting likelihood is not available in closed form and is computationally intractable to evaluate exactly, which makes simulation-based inference a natural approach [1, 5, 6].

However, the central issue in this problem is not computation alone. Even with a simulator in hand, inference can remain weak if the observed summaries do not contain enough information to distinguish transmission, recovery, and rewiring effects. For this reason, the report focuses on two linked questions: first, how summary-statistic design affects identification; and second, how different SBI methods perform once the summaries are fixed.

The observed dataset consists of $R = 40$ independent realizations generated from one common but unknown parameter vector $\theta = (\beta, \gamma, \rho)$. For each realization, three aggregate outputs are observed: the infected-fraction time series, the rewiring-count time series, and the final degree histogram. The main goals of the report are:

1. to explain why identification is difficult in this adaptive-network setting;
2. to show how richer summary statistics improve identification;
3. to compare several SBI methods on a common inferential task; and

4. to assess them using posterior predictive checks, calibration, and simulation budget rather than posterior concentration alone.

The main conclusion is that infected-only summaries identify at most a coarse net epidemic-growth balance and leave substantial ambiguity in ρ. Rewiring summaries are the key to identifying network adaptation, degree summaries provide additional structure, and regression-adjusted ABC on the full summary set gives the most convincing overall observed-data result in this implementation.

2 The Identification Problem in Adaptive-Network Epidemics

2.1 Why the likelihood is intractable

The simulator evolves a population of $N = 200$ agents on an undirected Erdős-Rényi graph $G(N, p)$ with $p = 0.05$, so the initial average degree is approximately 10 [5]. At time $t = 0$, five agents are infected and all others are susceptible. The process then runs for $T = 200$ discrete time steps. At each step, three synchronous updates are applied:

1. **Infection:** each susceptible neighbor of an infected node becomes infected with probability β;
2. **Recovery:** each previously infected node recovers with probability γ;
3. **Rewiring:** for each susceptible-infected edge, the susceptible endpoint cuts the edge with probability ρ and reconnects to a uniformly random non-neighbor.

The priors used in the analysis are

$$\beta \sim \mathrm{Unif}(0.05, 0.50), \qquad \gamma \sim \mathrm{Unif}(0.02, 0.20), \qquad \rho \sim \mathrm{Unif}(0, 0.80).$$

The likelihood is intractable because the data are aggregate outputs of a latent stochastic graph process. Exact evaluation would require integrating over an enormous space of hidden infection histories and hidden network trajectories. Rewiring makes the problem even harder because the network itself evolves in response to the epidemic, so the transition law is path-dependent. Even if one wrote down a complete-data likelihood for the fully observed graph process, marginalizing over all unobserved networks would be computationally infeasible. In practice, inference must therefore rely on the simulator rather than on a tractable likelihood [1, 3, 4, 6].

2.2 Why the parameters are hard to separate

A useful mean-field approximation writes the expected change in the infected fraction I_t as

$$\mathbb{E}[I_{t+1} - I_t \mid I_t, \lambda_t] \approx \beta\lambda_t - \gamma I_t,$$

where λ_t is the effective number of susceptible-infected contacts available for transmission. If the network were approximately homogeneous, one would have $\lambda_t \approx c_t I_t(1 - I_t)$ for some contact intensity c_t, so that

$$\mathbb{E}[I_{t+1} - I_t \mid I_t] \approx I_t\big[\beta c_t(1 - I_t) - \gamma\big].$$

Early in the epidemic, I_t is small, which reduces this to

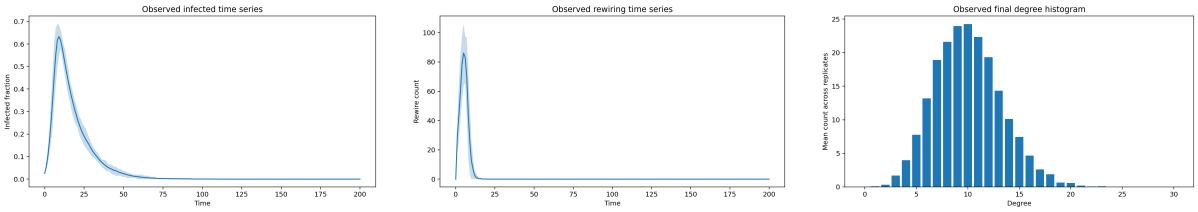
$$\mathbb{E}[I_{t+1} - I_t \mid I_t] \approx I_t(\beta c_t - \gamma).$$

This approximation makes the identification problem visible. Many pairs (β, γ) with similar value of $\beta c_t - \gamma$ can produce similar early epidemic growth. Rewiring complicates the problem

further because c_t is not fixed: a higher ρ removes susceptible-infected edges and reduces future transmission opportunities. As a result, different triples (β, γ, ρ) can generate very similar infected trajectories. This means that identification is fundamentally limited when inference is based only on infection summaries.

3 Why Summary Design Drives Identification

The observed data are summarized in figure 1. The infected trajectory has the familiar outbreak shape, the rewiring series peaks early and then decays quickly, and the final degree histogram records the cumulative network footprint of adaptation.



(a) Observed infected fraction. (b) Observed rewiring counts. (c) Observed final degree histogram.

Figure 1: Observed aggregate data channels used for inference. The infected trajectory alone does not reveal how many risky susceptible-infected edges were removed, which is why rewiring and degree information are informative for ρ.

For each realization, the pipeline extracts summary statistics and then averages them over the 40 replicates. Three nested summary sets are compared:

1. **infected only:** peak infected fraction, peak time, final infected fraction, mean infected fraction, infected AUC, early growth, mid growth, and time above 10%;
2. **infected + rewiring:** all epidemic summaries plus total rewires, peak rewires, mean rewires, early rewires, and late rewires;
3. **full:** all previous summaries plus mean degree, degree variance, and coarse degree-histogram shares.

Simulated and observed summaries are compared with the standardized Euclidean distance

$$d\left(s^{\text{sim}}, s^{\text{obs}}\right) = \left\| \frac{s^{\text{sim}} - s^{\text{obs}}}{\hat{\sigma}_s} \right\|_2,$$

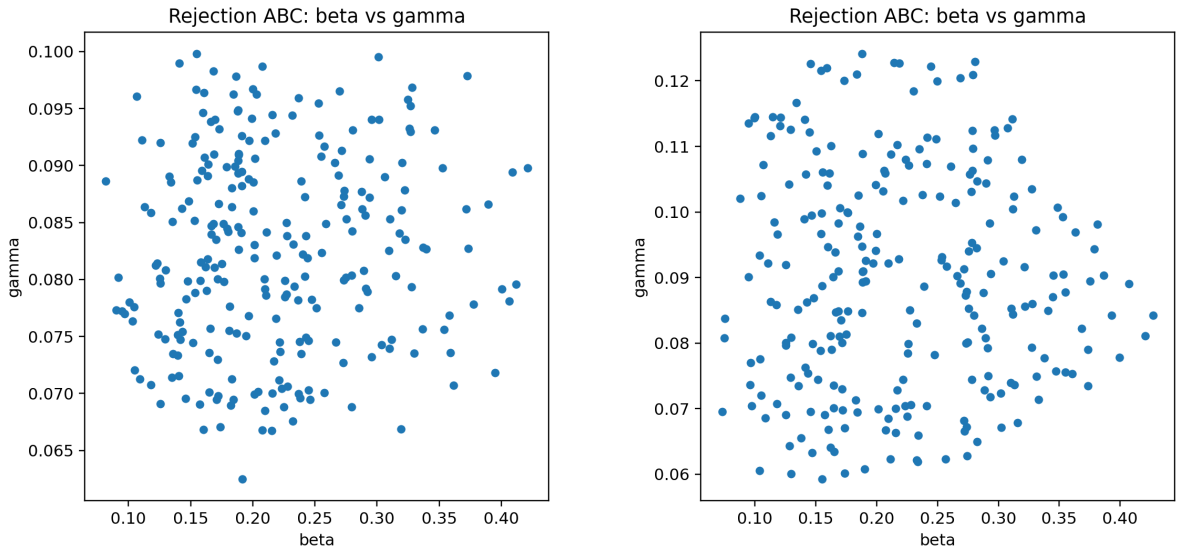
where $\hat{\sigma}_s$ is the across-replicate standard deviation of each observed summary component.

The baseline rejection-ABC experiment uses 5,000 prior draws and accepts the closest 5% according to this distance. The same simulated parameter draws are reused across the summary sets, so differences across summary sets reflect information content rather than Monte Carlo luck.

Table 1: Baseline comparison across summary sets using regression-adjusted ABC.

Summary set	$\hat{\beta}$	$\hat{\gamma}$	$\hat{\rho}$	Infected RMSE	Rewiring RMSE	Degree RMSE
infected only	0.178	0.081	0.351	0.0063	1.7145	0.4023
infected + rewiring	0.160	0.082	0.307	0.0042	0.6198	0.3250
full	0.155	0.082	0.302	0.0024	0.4197	0.3335

Table 1 gives the main empirical summary-design result of the report. Infected-only summaries produce weak identification of ρ and poor rewiring fit. Once rewiring summaries are added, the posterior mean of ρ moves from 0.351 to 0.307 and the rewiring RMSE falls sharply from 1.7145 to 0.6198. Adding degree information yields a smaller but still meaningful additional gain. This pattern indicates that the richer summaries are not merely reducing variance; they are adding missing information about network adaptation.



(a) Infected-only summaries.

(b) Full summary set.

Figure 2: Posterior geometry under rejection ABC. With infected-only summaries, the accepted draws form a broad ridge in the (β, γ) plane. The full summary set compresses this ridge substantially.

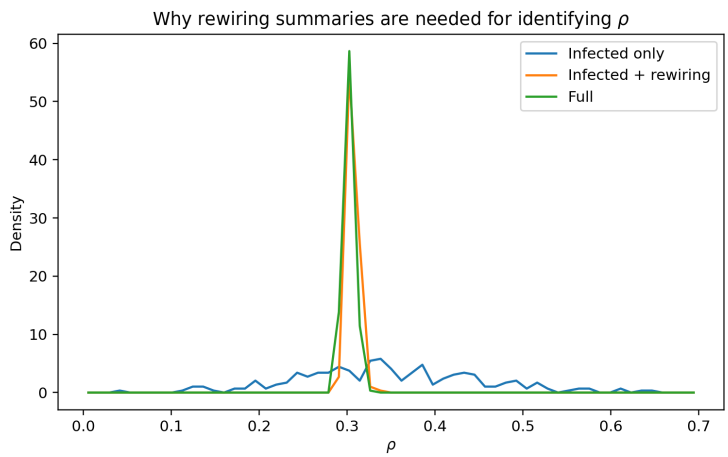


Figure 3: Posterior density for ρ across the three summary sets. Rewiring summaries are the main source of information for ρ, while degree summaries provide a smaller additional refinement.

Figures 2 and 3 reinforces the same message visually. Infected-only summaries leave a broad ridge in (β, γ) and a diffuse posterior for ρ, whereas the richer summary sets sharply improve identification. This is the central modelling lesson of the report: in this problem, summary design is at least as important as algorithm choice.

4 Inference Methods

Once the summary statistics are fixed, I compare five likelihood-free methods:

1. rejection ABC;
2. regression-adjusted ABC [1];
3. ABC-MCMC [3];
4. SMC-ABC [2, 4];
5. synthetic likelihood [6].

Regression adjustment post-processes accepted ABC draws by regressing parameters on the discrepancy between simulated and observed summaries and extrapolating to the observed summaries. In this project, it serves as the main benchmark because it improves on plain rejection ABC while remaining transparent and relatively easy to diagnose.

ABC-MCMC and SMC-ABC are included as standard sequential or chain-based refinements of likelihood-free inference. Synthetic likelihood takes a different approach: instead of accepting or rejecting draws by distance, it assumes that the chosen summary vector is approximately multivariate Gaussian conditional on θ and estimates that Gaussian distribution by repeated simulation at each parameter value.

To make the method comparison fairer, all advanced methods are evaluated on the same full summary set, and posterior predictive checks are used as the main criterion of practical performance.

5 Observed-Data Results

On the full summary set, the adjusted-ABC benchmark produces posterior mean

$$\hat{\theta}_{\text{adjABC}} = (\hat{\beta}, \hat{\gamma}, \hat{\rho}) = (0.155, 0.082, 0.302),$$

with 95% credible intervals $[0.136, 0.190]$, $[0.0788, 0.0870]$, and $[0.2925, 0.3148]$ respectively.

Table 2: Posterior means and 95% credible intervals on the full summary set.

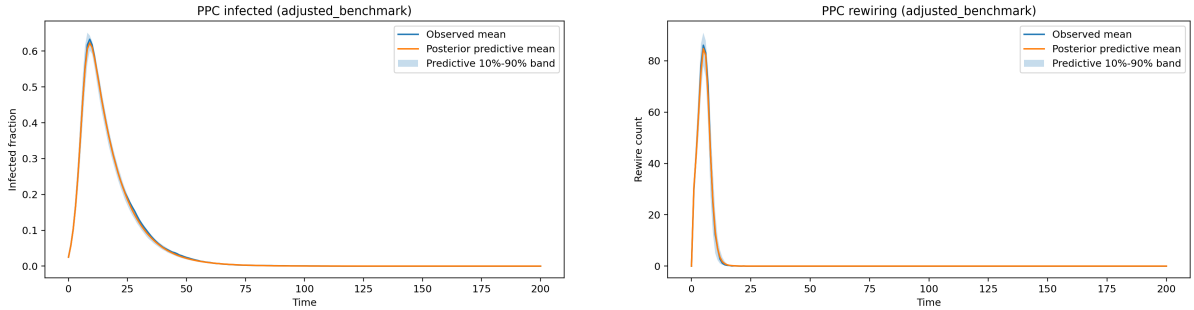
Method	Parameter	Mean	2.5%	97.5%
Adjusted ABC benchmark	β	0.155	0.136	0.190
	γ	0.082	0.079	0.087
	ρ	0.302	0.293	0.315
ABC-MCMC	β	0.196	0.085	0.334
	γ	0.086	0.061	0.119
	ρ	0.327	0.171	0.495
SMC-ABC	β	0.200	0.103	0.297
	γ	0.085	0.067	0.108
	ρ	0.348	0.213	0.460
Synthetic likelihood	β	0.158	0.152	0.160
	γ	0.080	0.079	0.080
	ρ	0.304	0.296	0.306

Two features stand out in table 2. First, ABC-MCMC and SMC-ABC remain much more diffuse than the adjusted-ABC benchmark in this implementation. Second, synthetic likelihood

produces posterior means close to adjusted ABC but dramatically narrower intervals. By itself, however, posterior sharpness is not enough to establish superiority. The next sections therefore turn to posterior predictive fit, calibration, and computational cost.

6 Posterior Predictive Evaluation

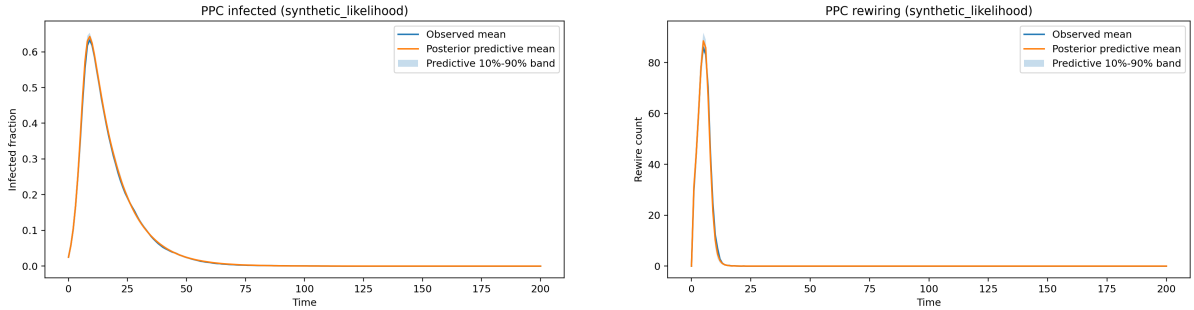
Posterior predictive checks ask whether the inferred posterior can reproduce the data that were actually observed. For each method, I generate posterior predictive simulations and compare them with the observed means of the infected trajectory, rewiring trajectory, and final degree histogram.



(a) Adjusted ABC: infected PPC.

(b) Adjusted ABC: rewiring PPC.

Figure 4: Posterior predictive checks for the adjusted-ABC benchmark. The posterior predictive mean tracks the observed mean closely, especially for the epidemic trajectory and the rewiring peak.



(a) Synthetic likelihood: infected PPC.

(b) Synthetic likelihood: rewiring PPC.

Figure 5: Posterior predictive checks for synthetic likelihood. The fit is also strong, but the rewiring PPC is slightly worse than the adjusted-ABC benchmark.

Table 3: Posterior predictive RMSE across advanced methods.

Method	Infected RMSE	Rewiring RMSE	Degree RMSE
Adjusted ABC benchmark	0.0023	0.4089	0.3606
ABC-MCMC	0.0207	4.3855	0.3248
SMC-ABC	0.0164	3.7086	0.3638
Synthetic likelihood	0.0028	0.5274	0.3353

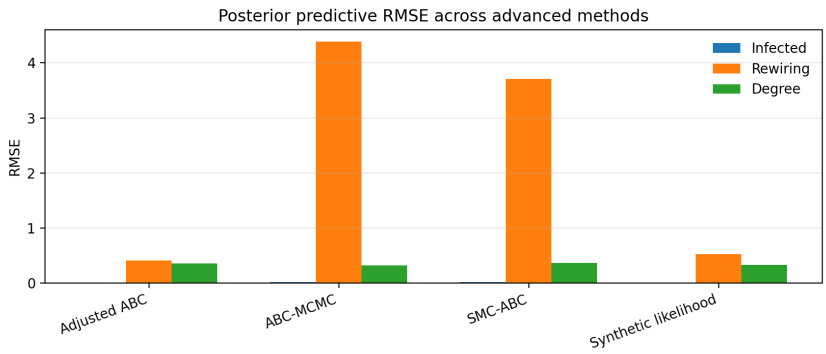


Figure 6: Posterior predictive RMSE comparison across advanced methods. Adjusted ABC is the strongest overall method in this run; synthetic likelihood is competitive but not uniformly better.

The evidence in table 3 and figure 6 supports regression-adjusted ABC as the strongest overall observed-data result in this implementation. Synthetic likelihood is competitive on infected and degree fit but somewhat worse on rewiring. ABC-MCMC and SMC-ABC remain too diffuse to reproduce the rewiring dynamics accurately. This is why the report treats adjusted ABC, rather than the visually sharpest posterior, as the most convincing main result.

7 Synthetic Likelihood: Efficiency vs. Overconfidence

Synthetic likelihood assumes approximately Gaussian summaries conditional on θ, which may be imperfect in this model due to skewness or multimodality. This helps explain the much tighter posteriors in figure 7, especially for β and ρ. While this may reflect efficiency, it may also indicate underestimated uncertainty when the Gaussian assumption is violated.

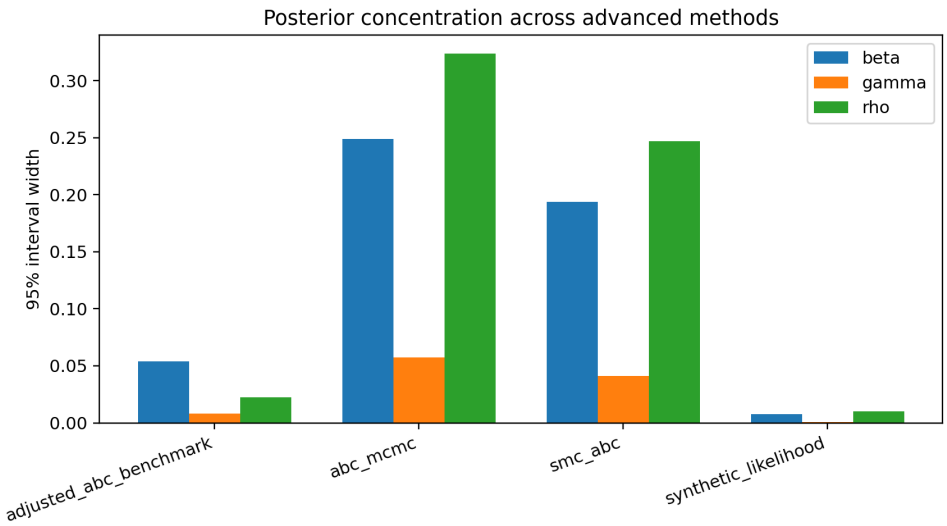


Figure 7: 95% interval widths across advanced methods. Synthetic likelihood is dramatically tighter than the ABC-based methods. This may indicate efficiency, but it may also indicate underestimation of uncertainty when the Gaussian summary assumption is imperfect.

8 Calibration via Synthetic-Truth Recovery

A limitation of the observed-data analysis is that the true parameter vector is unknown. To assess calibration more directly, I added one synthetic-truth recovery run. I chose

$$\theta_{\text{true}} = (\beta, \gamma, \rho) = (0.18, 0.08, 0.30),$$

generated a synthetic dataset from this parameter, and reran the inference pipeline on the synthetic data.

Table 4: Synthetic-truth recovery results. All three true parameters are covered by the reported 95% credible intervals in this run.

Method	Parameter	True value	Posterior mean	2.5%	97.5%
Adjusted ABC	β	0.180	0.193	0.173	0.226
	γ	0.080	0.083	0.080	0.087
	ρ	0.300	0.303	0.295	0.315
Synthetic likelihood	β	0.180	0.182	0.178	0.186
	γ	0.080	0.081	0.079	0.082
	ρ	0.300	0.304	0.299	0.312

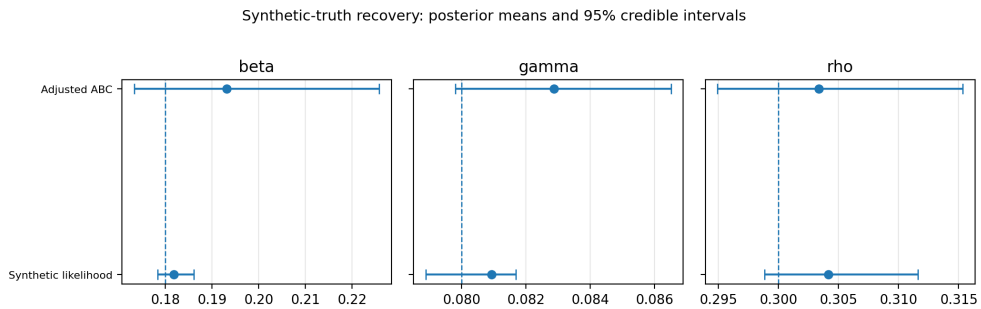


Figure 8: Synthetic-truth recovery: posterior means and 95% credible intervals for the two strongest methods. In this run, both methods cover the chosen true value for all three parameters.

The recovery experiment gives two useful messages. First, the narrow synthetic-likelihood intervals are not obviously miscalibrated in this particular run, because they do cover the chosen truth. Second, adjusted ABC is also calibrated and remains more conservative in its uncertainty reduction. The correct conclusion is therefore not that synthetic likelihood is invalid, but that its sharpness should be interpreted alongside its assumptions and computational cost.

9 The Role of Simulation Budget

A fair method comparison should distinguish methodological quality from raw computational budget. Synthetic likelihood often requires many internal simulator calls because each likelihood evaluation must itself be estimated by repeated simulation at a fixed parameter value.

Table 5: Explicit simulation budget from the add-on analysis.

Method/run	Simulator calls	Total replicate simulations
Rejection ABC (actual data)	2,500	100,000
Rejection ABC (synthetic truth)	2,500	100,000
Synthetic likelihood (actual data)	19,216	768,640
Synthetic likelihood (synthetic truth)	19,216	768,640

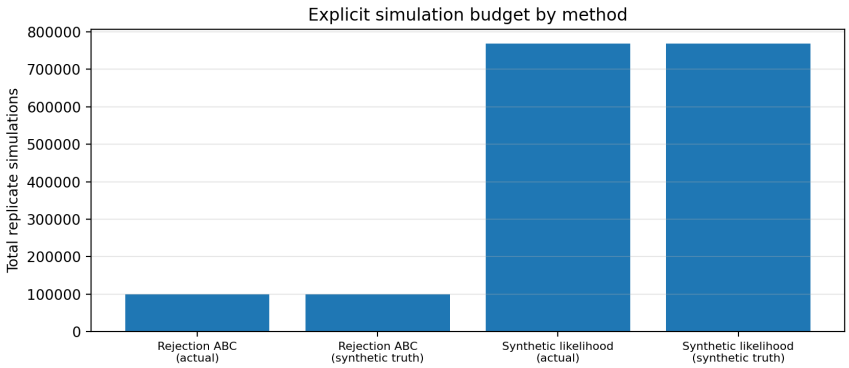


Figure 9: Explicit simulation budget by method. Synthetic likelihood uses roughly 7.7 times as many total replicate simulations as the rejection-ABC runs in the add-on analysis.

Table 5 and figure 9 makes an important point transparent: synthetic likelihood is not operating at the same computational budget as the rejection-ABC baseline. Therefore, its very tight posterior should not be interpreted purely as a methodological gain. Part of the apparent improvement may simply come from using much more simulation effort.

10 Discussion: What Actually Matters in SBI

The results of this project suggest three broader lessons. First, in adaptive-network epidemic models, identification is the primary inferential challenge. If the summaries do not distinguish transmission, recovery, and rewiring effects, even sophisticated methods cannot recover that missing information.

Second, summary design matters at least as much as method choice. The clearest improvement in this project comes not from moving from one algorithm to another, but from expanding the observed summaries beyond infected trajectories to include rewiring counts and degree information.

Third, posterior sharpness is not a sufficient measure of inferential quality. A convincing SBI result should combine four ingredients: good posterior predictive fit, reasonable calibration, robustness to modelling assumptions, and transparent computational cost. Under that broader standard, regression-adjusted ABC is the strongest overall result here, while synthetic likelihood is competitive but should be interpreted more cautiously.

11 Transparency Statement

AI tools were used to help structure code, identify implementation bugs, and improve the clarity of English explanations. However, the final model choices, mathematical interpretations, simulator settings, and reported conclusions were checked manually against the simulator behaviour and the empirical output. Two non-final attempts are especially relevant. First, infected summaries

alone led to weak identification and did not pin down rewiring behaviour well. Second, synthetic likelihood produced very tight posteriors, but these were not accepted at face value; I therefore added posterior predictive checks, a synthetic-truth recovery exercise, and an explicit simulation-budget comparison before interpreting the method as competitive. My own contribution was to build the unified comparison pipeline, keep the baseline ABC comparison fair through shared simulations, quantify the remaining identification issues, and interpret the results in terms of epidemic mechanisms rather than only reporting numerical output.

12 Conclusion

This project shows how careful simulation-based inference can be carried out even when the likelihood is unavailable in practice. The main substantive result is that infected-only summaries are not enough to identify transmission, recovery, and rewiring separately. Rewiring summaries are essential for identifying ρ, and degree summaries provide useful additional structure.

Among the methods implemented here, regression-adjusted ABC with the full summary set gives the strongest overall observed-data result because it combines good posterior predictive performance with transparent assumptions and a moderate computational burden. Synthetic likelihood is also competitive and passes the added recovery check in this run, but its much tighter posterior should still be interpreted cautiously because it relies on stronger assumptions and substantially greater simulation effort.

More broadly, the project demonstrates that in likelihood-intractable epidemic models, good inference depends not only on algorithmic sophistication but on choosing summaries that match the scientific mechanisms of the model. In this sense, the most important design decision in SBI is often not the method itself, but what information is retained from the data.

GitHub repository. https://github.com/kodai1126/Simulation-Based-Inference-for-an-Adaptive-Network-Epidemic-Model

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